Karst relief denudation based on limestone tablets weight loss (Slovak karst)

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Introduction

Currently, chemical denudation is one of the main geomorphological factors. Various methodologies are used to monitor it over time, mostly indirectly based on the detection of karst water mineralization. Direct methods are rarely used, one of the possibilities is based on accurate measurement of weight loss on limestone plates (samples).

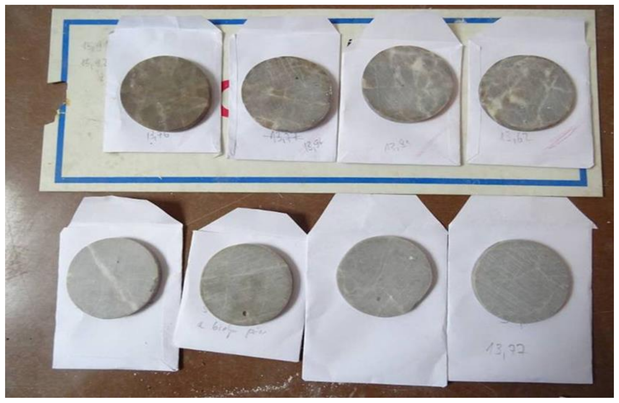
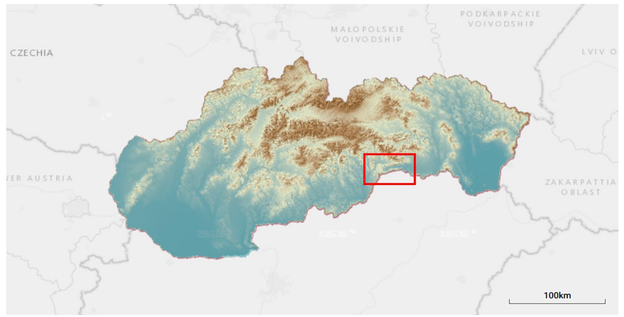


Fig. 1 The first generation of plates made of local material
Source: Hochmuth, Z. (2018)

Researched areas in the Slovak Karst

- Silica plateau (Silica village)
- Jasov plateau (near the village Háj)

The aim of our work is based on measurable results to analyze the degree of denudation and its regularity in the localities of the Slovak Karst.



Map . 1 Study area – Slovak karst

Methods

Our research is based on method of Ivan Gams and measurable results of limestone tablets weight loss at two experimental sites of the Slovak Karst. On Silica and Jasov Plateau was monitored weight loss in the 3 months intervals from December 2016 until present. We used two different types of tablets (from the Lipica quarry in Slovenia and from Slovak karst since 2018) in three different possitions – one the surface and at a depth of 20 and 50 cm. The dissociation rate (mg/year/cm^2) was calculated based on this equation:

$$DR = (W_1 - W_2) * \frac{1000}{t} * \frac{365}{S}$$

Fig. 2 Placement of Chinese samples on Jasovská plateau on the surface
Source: Hochmuth, Z., (2018)

Fig. 3 Placement of Chinese samples on the Jasov plateau at a depth of 20 cm
Source: Hochmuth, Z., (2018)

Weight loss monitoring

During the monitoring of this process, we used a method based on the transmission of weight loss measurements using KERN ABT-NM analytical balances together with the study and measurement of various factors that affect it.

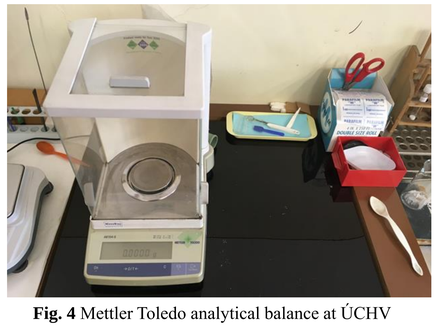


Fig. 4 Mettler Toledo analytical balance at ÚCHV

		01-03/2017	04-06/2017	07-09/2017	10-12/2017	01-03/2018	04-06/2018	07-09/2018	10-12/2018	01-03/2019	04-06/2019	07-09/2019	10-12/2019	01-03/2020	04-06/2020	07-09/2020	10-12/2020
Standard tablets	J_50	5,87304678	8,34329453	4,80017598	2,18308184	14,1400958	7,31613957	4,7734007	0,63659982	4,74581722	3,64579008	1,72296485	4,98667249	6,73541813	3,42734204	3,12348948	6,51878100
	J_20	5,16027727	7,70702264	5,83728819	5,30363024	6,07540070	4,02517613	3,03484289	0,91087344	1,75826039	3,22934361	1,68702543	5,49422221	6,23681586	4,67902204	4,50390289	5,27254346
	J_0	0,61273085	0,79413438	1,16315229	0,36023022	0,31725986	1,47195500	0,86803039	0,45603220	0,78304449	0,86724333	0,73187822	0,79671359	2,70183226	0,63364573	1,49791643	1,96539815
	S_50	4,37248050	7,96057460	4,49536737	2,43048865	5,69489940	5,25631244	4,67393563	1,51004275	3,67003902	2,80474176	1,97157166	4,72158637	2,71200282	1,88727028	1,49791643	4,37024082
	S_20	2,76790929	7,05161476	6,08651225	3,79326745	12,0153063	5,25631244	8,31482185	3,31392937	2,48137594	4,88447494	3,16364353	4,12479562	1,83063586	2,18915801	1,37207153	7,43431164
	S_0	0,30428129	0,67454388	0,73782890	0,08875979	0,18872051	0,62286516	0,81422614	1,25532138	0	0	0,66295149	0,84892328	0,28156684	0,83486732	0,62605573	1,30456775
Local tablets	J_50								1,47746	3,46804	1,88791	1,06548	6,04456	2,85291	3,91618	2,47189	5,43881
	J_20								1,40523	2,06876	2,77536	2,65776	3,54806	1,76712	2,95608	3,84236	5,62451
	J_0								1,07759	0,70650	0,76239	0,32851	0,83319	0,58716	0,87818	1,07759	0,70650
	S_50								1,52468	3,00557	4,97942	3,02354	3,62611	1,94028	1,27759	1,16738	4,12149

Tab.1 Overview of individual dissolution rates of all plates for the observed period

Results

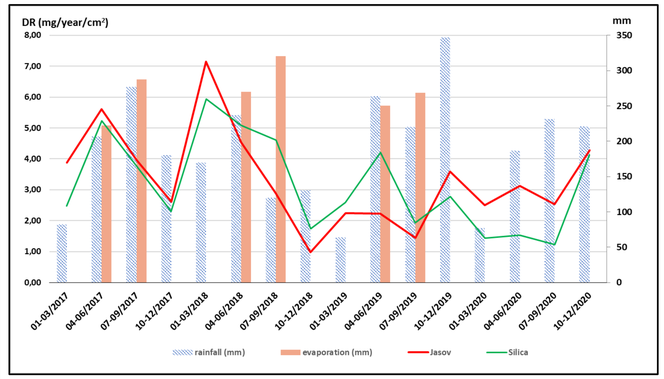


Fig. 5 The graph shows the dependence of rainfall (mm), evaporation (mm) and the values of the dissolution rate of the original plates on the Silica and Jasov plateau

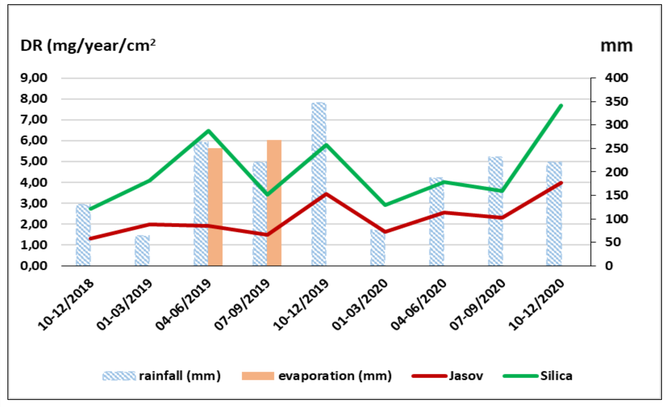


Fig. 6 The graph shows the dependence of rainfall (mm), evaporation (mm) and the values of the dissolution rate of local plates from Silická and Jasovská planina

Conclusions

There are no significant differences in denudation rate (DR) between experimental sites. Lowest DR was measured on the surface .The highest DR was recorded in the Jasov Plateau at a depth of 50 cm by 3.40%, but on the Silica Plateau at a depth of 20 cm and 3.62% (at a depth of 50 cm it was 2.62%).The most significant factors was rainfall and evapration; temperature and biological (organic) activity too.

References

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